

Dear Editors,

Please consider our manuscript entitled “Habitat coupling by small fish across lake–creek habitats: a missing link in freshwater food webs?” for publication as a Letter in *Ecology Letters*.

Habitat coupling is a central concept in food web ecology, yet it is often inferred from the movements and resource use of large-bodied consumers. In this manuscript, we show that small-bodied fishes can also act as important habitat couplers, linking adjacent freshwater lake and creek habitats through both resource assimilation and movement behaviour. Using a multi-method field approach including stable isotope analysis, acoustic telemetry, and seasonal biomass flux estimates, we demonstrate that these fishes integrate carbon from both habitats, move frequently across the lake–creek boundary, and contribute to seasonal exchange of biomass between habitats.

Our results reveal that coupling is not constant through time, but instead varies across seasons, with the strongest lake–creek habitat coupling occurring in spring and fall and the weakest in summer. This seasonal pattern suggests that small-bodied consumers may play an underappreciated role in maintaining energy flows across spatially distinct habitats, particularly when environmental conditions or predation risk vary through time. In doing so, our study advances understanding of how spatially structured food webs are connected and highlights the need to look beyond large predators when evaluating habitat coupling and the processes that promote resilience in freshwater systems.

We believe this manuscript will be of broad interest to *Ecology Letters* readers because it addresses a general ecological question with clear implications for food web theory and spatial ecology. It also speaks to how movement, resource use, and biomass exchange jointly shape the functioning of connected ecosystems and promises to excite further research on habitat coupling across taxa and temporal scales. Further, our findings have important implications for managing habitat connectivity and protecting the ecological processes that maintain freshwater biodiversity in the face of ongoing global change.

We have taken care to ensure the manuscript is focused, original, and appropriate for the scope and format of a Letter. Thank you for your consideration.

Sincerely,



Dr. Charlotte A. Ward  
on behalf of all co-authors

**Title:** Habitat coupling by small fish across lake-creek habitats: a missing link in freshwater food webs?

**Authors:** Charlotte A. Ward<sup>1,2\*</sup>, Reilly F. O'Connor<sup>1</sup>, Timothy J. Fernandes<sup>1,2</sup>, Kevin S. McCann<sup>1,2</sup>, Mark S. Ridgway<sup>3</sup>, Bailey C. McMeans<sup>4</sup>, Tyler D. Tunney<sup>5,1</sup>

**Affiliations:**

1. University of Guelph; Guelph, Canada
2. Centre for Ecosystem Management; Guelph, Canada
3. Ontario Ministry of Natural Resources; Peterborough, Canada
4. University of Toronto Mississauga; Mississauga, Canada
5. Fisheries and Oceans Canada; Moncton, Canada

\*Corresponding author: Charlotte A. Ward; Address: 50 Stone Road East, Guelph, ON Canada, N1G 2W1; Orchid ID: 0000-0001-7982-1239; Email: [cward@uoguelph.ca](mailto:cward@uoguelph.ca)

**Article Type:** Letter

**Main Text Word Count:** 4979

**Abstract Word Count:** 175

**Number of Figures:** 6

**Number of Tables:** 0

**Novelty statement:**

This manuscript presents novel evidence that small-bodied fishes can function as important habitat couplers across adjacent freshwater lake and creek habitats. By integrating stable isotope analysis, acoustic telemetry, and seasonal biomass flux estimates, we demonstrate that lake–creek coupling is temporally dynamic, with coupling strength varying across seasons. These findings advance our understanding of spatial food web linkages by broadening the range of organisms recognized as habitat couplers and highlight the importance of considering small-bodied consumers in studies of spatial connectivity and freshwater ecosystem function and resilience.

**Authorship Statement:**

Charlotte A. Ward conceived the study with input from Tyler D. Tunney and Kevin S. McCann. Charlotte A. Ward led the study design, field data collection, data curation, statistical analyses, interpretation of results, and writing of the original manuscript. Reilly F. O'Connor, Timothy J. Fernandes, Mark S. Ridgway, Bailey C. McMeans, Kevin S. McCann, and Tyler D. Tunney contributed to study development, field sampling, methodological development, interpretation of results, and manuscript revision. Kevin S. McCann and Tyler D. Tunney supervised the project. All authors provided critical feedback, approved the final version of the manuscript.

**Data availability statement:**

All data and code necessary to reproduce the analyses presented in this manuscript are currently available for review via a public GitHub Repository linked [here](#). This repository includes data and code for stable isotope analysis, acoustic telemetry-derived movement summaries, and seasonal biomass flux estimates. Repository details and a permanent DOI will be provided in the final published version of the article.